

ITC C103
WW2 - M3U5_M4U1 Midterms Assignment 10

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Answers and Questions:

M3U5 (File Systems)

1. Logical vs. Physical Drives: In Windows, you see your storage as letter-labeled "drives" (e.g., C:, D:). Explain the difference between these logical drives and the physical hard disks inside a computer. Why might a single physical disk have multiple logical drives, or a single logical drive represent more than one physical device?

Answer: A physical drive is the actual hardware, like an HDD or SSD. A logical drive is a partition that Windows shows with a letter, like C: or D:. One disk can be split into several logical drives. One logical drive can also span several disks, as in RAID.

2. File Organization and Paths: If you were an IT administrator trying to locate a user's personal files, how would you find the default location for their documents, and why is it important not to move or delete essential system folders like C:\Windows?

Answer: Documents are found in C:\Users\username\Documents. C:\Windows holds the system files, so moving or deleting them can stop Windows from working.

3. File Handling and Safety: You're working on a crucial project and need to save a new version of a file without overwriting the original. What file command would you use? Additionally, why is it safer to delete files from your main hard drive compared to deleting them from a USB flash drive?

Answer: Use Save As to keep the original. Files deleted from the main drive go to the Recycle Bin and can be restored. Files deleted from a USB drive are gone permanently.

4. File Search and Management: Imagine you have a large folder full of mixed documents, photos, and videos. Describe the steps you would take within File Explorer to find all the image files from last month and sort them by size, demonstrating a practical understanding of search and view options.

Answer: Open the folder, then use the search box to filter by Kind: Picture. Filter by Date modified: Last month. Switch to Details view and click the Size column to sort.

5. File Formats and Compression: Your friend wants you to send them a high-quality photo from your phone, but they're concerned about the file size. Explain the difference between a high-quality image format like TIFF and a common compressed format like JPEG. How would using a ZIP file help you send multiple files efficiently, and what is the difference between a compressed file and a regular folder?

Answer: TIFF keeps full quality but is very large. JPEG is much smaller with a small loss in quality. A ZIP file bundles files into one smaller archive, while a regular folder only organizes files and does not shrink them.

M4U1 (Networking Concepts)

1. Network Hierarchy: The internet is often described as a "network of networks." Explain the key differences between a Local Area Network (LAN) and a Wide Area Network (WAN). How do these two concepts relate to each other in connecting a home network to the internet?

Answer: A LAN covers a small area like a home or office. A WAN covers a large area and connects many LANs. A home LAN reaches the internet, the biggest WAN, through a router.

2. Packet Switching and Routers: When you send an email or visit a website, the information is broken down into small data packets. Explain the purpose of a packet in data transmission. What is the role of a router in this process, and what information does it use to determine the best path for a packet to reach its destination?

Answer: A packet is a small piece of data sent across a network. A router forwards packets toward their destination, using the destination IP address to choose the best path.

3. Addressing and Protocols: When you type a website name like `www.google.com` into a browser, your computer needs to find the correct server to connect to. Describe the function of both an IP address and a MAC address in this process. What is the role of DNS (Domain Name System) in resolving the human-readable website name to a numerical IP address?

Answer: An IP address identifies a device on a network and is used to route data between networks. A MAC address is the hardware address of a network card and is used inside the local network. DNS converts a website name into its IP address.

4. Email Protocols: You're setting up an email account on your phone and notice you have two options for receiving messages: POP3 and IMAP. Explain the difference between these two protocols. What are the practical implications of choosing one over the other for a user who checks their email on multiple devices?

Answer: POP3 downloads emails to one device and usually removes them from the server. IMAP keeps emails on the server and syncs them across devices, so it is better for users with multiple devices.

5. Web Security: When you visit a website with `https://` at the beginning of its URL, it means the connection is secure. Explain what this means from a technical standpoint, and what SSL/TLS and a digital certificate do to protect your information, such as passwords or credit card numbers, from being intercepted.

Answer: HTTPS means the connection is encrypted by SSL/TLS, so intercepted data is unreadable. A digital certificate proves the website is genuine and provides the key for the encryption. This protects passwords and card numbers.